EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

Patient Appointment Scheduling System — Security and Governance Use Case

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1. Aim

To plan and apply security features and governance and administration practices in the Patient Appointment Scheduling System built in Zoho Creator — covering sign-in security, user access, data permissions, safe scripting and integrations, controlled changes and auditing — so that patient data is protected and the application is managed in an orderly way.

The application stores personal and medical details such as age, existing conditions, symptoms and consent, so access control and accountability are essential.

2. Procedure

Step 1: Identify what needs protection — The Patient Profile, Appointment Records and Departments forms were reviewed, and the sensitive data was listed: patient identity, contact details, medical conditions, symptoms and treatment consent.

Step 2: Plan account security — Sign-in protection was planned for all users: strong passwords, two-factor sign-in and single sign-on where the organisation uses it (Section 3.1).

Step 3: Design roles and permissions — Four user types — Administrator, Doctor, Front desk and Patient — were defined, and a permission matrix was prepared following the principle of least privilege (Section 3.2).

Step 4: Define the access flow — The checks made when a user opens the application were drawn as a decision flow (Section 3.3).

Step 5: Secure scripts and connections — Safe Deluge practices were drafted, such as validating input and keeping connections limited to administrators, based on the connection and workflow work in Experiments 6 and 7 (Section 3.4).

Step 6: Set up governance — Controlled changes through development, stage and production environments, audit review and periodic access reviews were planned (Section 3.5).

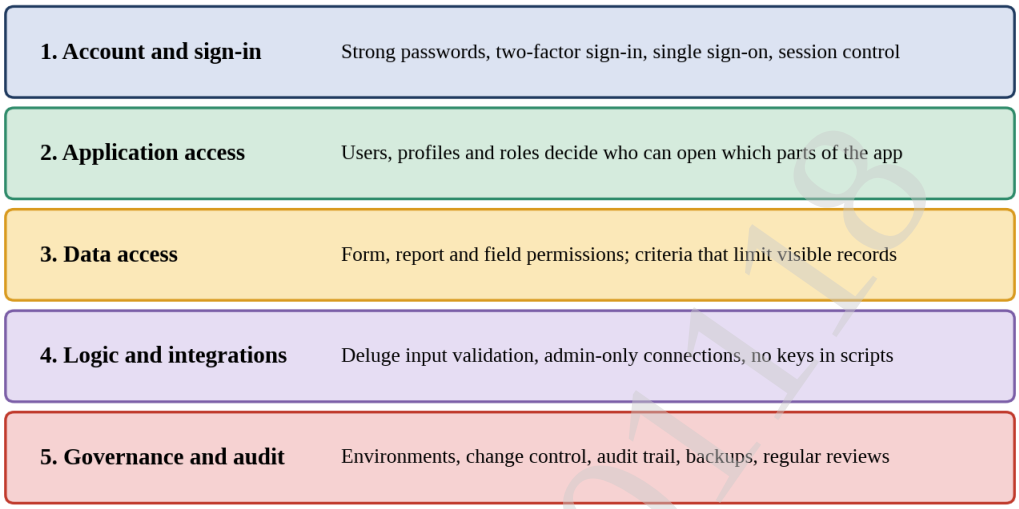
Step 7: Prepare the admin checklist — Threats and matching controls were tabulated, and an administration checklist was prepared for regular use (Sections 3.6 and 3.7).

Note: the settings are described as a planned configuration. No screenshots of the settings were captured for this record, so it does not claim that every option was turned on. Feature names and availability depend on the Zoho Creator plan and version and should be confirmed in the current documentation.

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3. Output

3.1 Layered security and governance model



Security is arranged in five layers: account sign-in, application access, data access, logic and integrations, and governance with audit. A failure in one layer is caught by the next, which is the idea of defence in depth.

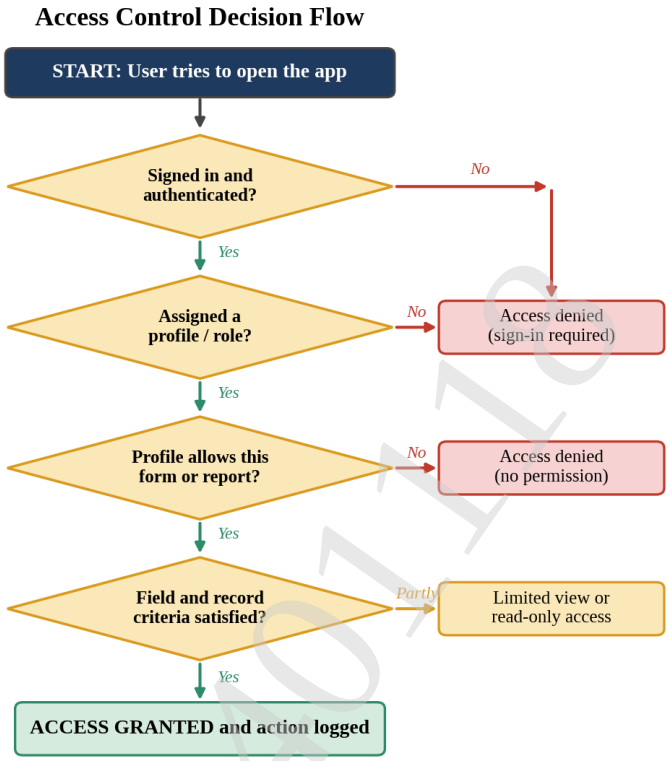
3.2 Proposed permission matrix

Each user type receives only the access needed for the job (least privilege). The matrix is a proposed design for the appointment system.

Form / report	Administrator	Doctor	Front desk	Patient
Patient Profile	View, edit, delete	View (own patients)	Create, view, edit	Create and view own record
Appointment Records	Full access	View and update status	Create, view, edit	View own appointments
Departments (master data)	Full access	View	View	No access
Medical fields (conditions, symptoms)	View, edit	View	Limited or hidden	Own entry only
Reports and exports	All reports, export	Own reports	Daily schedule only	No access
Users, connections, environments	Manage	No access	No access	No access

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3.3 Access control decision flow



When a user tries to open the application, the system checks sign-in, then the assigned profile or role, then permission on the form or report, and finally field and record criteria. Failing any check denies or limits access, and granted actions are logged.

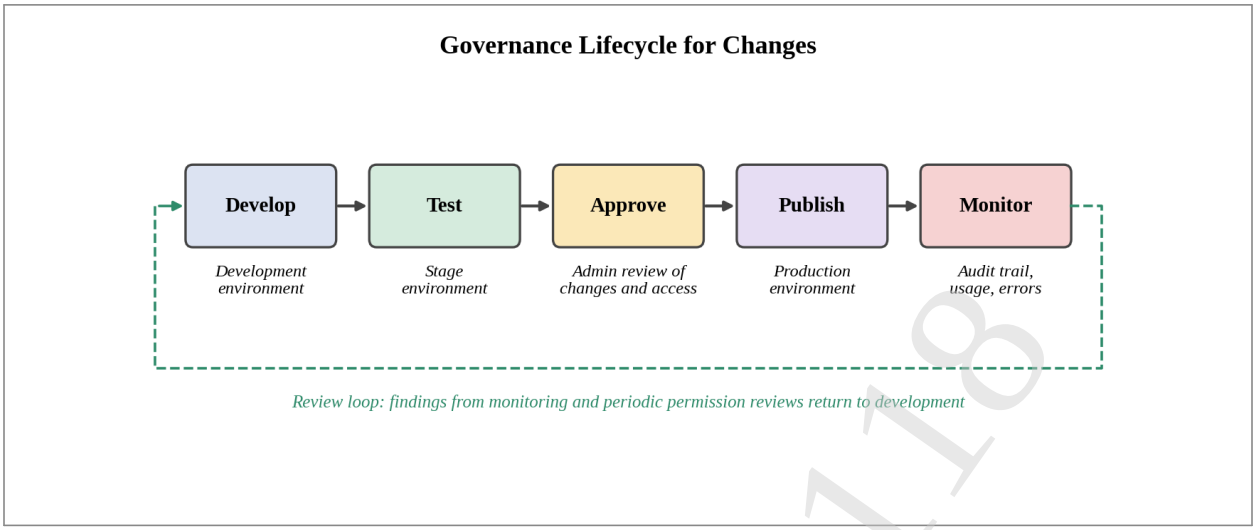
3.4 Secure scripting and connection practices (illustrative)

```
// On validate of the Patient Profile form
if(input.consent_for_treatment != true)
{
    alert "Treatment consent is required before submitting.";
    cancel submit;
}
if(input.age < 0 || input.age > 120)
{
    alert "Please enter a valid age.";
    cancel submit;
}

// Connection: use the stored link name, never a key written in the script
response = zoho.calendar.createEvent(calendar_id, event, "zoho_calendar_connection1");
```

Input is validated on the server side of the form so wrong or missing data is rejected. Credentials stay in the connection, which is set to administrators only (as in Experiment 7), and the calendar ID should be stored in a setting instead of being written inside the script. The code is illustrative.

3.5 Governance lifecycle for changes



Changes move from the development environment to the stage environment for testing, then to an administrator review, and only then to the production environment. Monitoring through the audit trail and periodic permission reviews feed improvements back into development.

3.6 Threats and matching controls

Threat	Example in this application	Control
Unauthorised access	A stranger or former staff member opens patient data.	Two-factor sign-in, remove users promptly, profile-based access.
Excess privilege	Front desk staff can delete or export all records.	Least privilege, separate permissions for delete and export.
Data leakage	Reports or forms shared publicly by mistake.	Review sharing settings, avoid public links for sensitive forms.
Insecure scripts	Missing validation or keys written inside Deluge code.	Validation on submit, connections instead of keys, code review.
Uncontrolled change	A live fix breaks booking or permissions.	Development, stage and production environments with approval.
No traceability	Nobody can tell who changed a record.	Audit trail review and periodic log checks.

3.7 Administration checklist

Area	Administration practice	How often
Users	Add users with the correct profile; remove leavers the same day.	On every change
Permissions	Review profiles and exports against the permission matrix.	Every quarter
Sign-in	Confirm two-factor sign-in or single sign-on is enforced.	Every quarter
Connections	Check that connections are administrator-only and used only by needed apps.	Every quarter
Changes	Publish only after testing in the stage environment.	Every release
Audit	Look through the audit trail and usage metrics for unusual activity.	Monthly
Backup and recovery	Keep a recent backup and test that data can be restored.	As planned

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4. Final Result

A layered security and governance plan was prepared for the Patient Appointment Scheduling System, covering sign-in protection, role-based access with a least-privilege permission matrix, an access-control decision flow, secure scripting and connection practices, controlled changes through environments, and regular auditing.

Threats such as unauthorised access, excess privilege, data leakage, insecure scripts, uncontrolled change and lack of traceability were matched with controls, and an administration checklist was drawn up for regular use.

The exercise met its objective of designing security and governance for the application. The settings were not captured as screenshots, so the record presents them as a planned configuration to be confirmed in the live application.

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